Statistical Machine Learning

M.Sc. in Data Science — Academic Year 2025-2026

Investigating the Double Descent Phenomenon

Factorization Models on Sparse E-Commerce Transaction Data

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1 Introduction

Recommender systems are usually trained from sparse and partially observed user–item matrices. In many real applications, and especially in e-commerce, the observed data do not correspond to explicit ratings but to implicit feedback: a purchase indicates interest, while the absence of a purchase does not necessarily indicate dislike. We study transaction data from the online store of the Chilean branch of the Mexican publishing house Fondo de Cultura Económica, whose catalogue includes books from a broad range of genres and subject areas. Since customers may also buy through physical and other online bookstores, unobserved online interactions must be interpreted as missing preferences rather than true negative labels.

This project aims to be an extension of the study presented in the paper 'Investigating Overparametrization for Non-Negative Matrix Factorization in Collaborative Filtering'[3]. Kawakami and Sugiyama showed that overparameterized Non-Negative Matrix Factorization (NMF) exhibits *double descent* in test RMSE as the number of latent factors k grows well beyond $\min(n, m)$: the error first decreases, rises near the interpolation threshold, and decreases again, with the overparameterized regime eventually outperforming the underparameterized one. Their experiments use explicit ratings on a 1–5 scale (MovieLens 100K, Douban) and measure reconstruction error on the observed entries.

This project, instead, studies collaborative filtering in the sparse implicit-feedback environment. We use matrix factorization models to learn latent representations of users and items and evaluate whether these representations can recover held-out purchases. Beyond recommendation accuracy, our main objective is methodological: to analyze how generalization changes as the capacity of matrix factorization models increases.

Matrix factorization provides a natural framework for studying this question. The latent dimension k controls model capacity directly, so a single sweep over k traces the model from the underparameterized to the overparameterized regime. We study three formulations that differ precisely in how they treat the unobserved cells:

- **Non-Negative Matrix Factorization (NMF)**, which reconstructs the binary matrix on a fixed sampled support Ω built from the positives plus r sampled negatives per positive.

- **Weighted Matrix Factorization (WMF)** (implicit ALS, iALS), the canonical implicit-feedback model, which fits every cell of the $n \times m$ matrix and distin-

guishes positives from unobserved cells through a confidence weight rather than through sampling. Note that, unlike NMF, iALS imposes no non-negativity constraint.

- **Logistic Matrix Factorization**, which abandons reconstruction altogether and models the probability of interaction with a logistic loss and negative sampling.

Together, these three models span the two opposite treatments of the unobserved entries (sampled support versus the full matrix) and the two loss families (squared reconstruction versus probabilistic binary classification). Contrasting them allows us to evaluate whether double-descent-like behavior appears in a real bookstore recommendation problem and to isolate which modeling ingredient the phenomenon actually depends on.

2 Theoretical Background and Research Motivation

A central question in statistical learning is how model complexity affects generalization. In the classical bias-variance framework, increasing model complexity initially improves performance by reducing bias, but after a certain point, it increases variance and worsens test performance. Under this view, very large models are expected to overfit, and controlling complexity is necessary to achieve good generalization.

However, modern machine learning has challenged this classical intuition. Highly overparameterized models can often interpolate the training data and still generalize well. This empirical behavior is summarized by the *double descent* phenomenon: as model complexity increases, test error first decreases, then increases near the interpolation threshold, and finally decreases again in the overparameterized regime. In this sense, the classical U-shaped test error curve is followed by a second descent once the model becomes sufficiently large.

Traditionally, matrix factorization models are used in an underparameterized regime, where the number of latent factors is much smaller than the number of users or items. This is consistent with the classical low-rank assumption: user preferences are assumed to be explained by a relatively small number of latent dimensions. However, if double descent occurs in collaborative filtering, then this conventional restriction may be too conservative. Larger latent representations may initially overfit but later improve test performance in the overparameterized regime.

Kawakami and Sugiyama explicitly investigate this issue for non-negative matrix factorization in collaborative filtering. Their empirical results show that overparameterized NMF can outperform underparameterized NMF and can be competitive with more complex recommender models. More importantly for our project, they observe a double-descent-like pattern in test RMSE when the number of latent factors increases. This suggests that the relationship between latent dimension and generalization in recommender systems may be more complex than predicted by the classical bias-variance trade-off.

The objective of this project is not only to build a recommender system, but to empirically study how matrix factorization models behave as complexity increases in a sparse implicit-feedback environment.

More specifically, we evaluate the following questions:

1. How does recommendation performance change as the latent dimension k increases?

2. How does training time affect generalization, particularly when the number of epochs increases?

3. Does the phenomenon appear to the same extent in a pointwise metric and a ranking metric? Or does it appear in one but not in the other?

4. Does the phenomenon depend on how the negatives are treated?

5. What is the effect of the regularization?

By answering these questions, the project connects three elements: the theory of double descent, matrix factorization for collaborative filtering, and real-world e-commerce data. This provides a concrete empirical setting to analyze whether overparameterization can improve recommendation performance beyond the classical low-rank regime.

3 Dataset

3.1 Data source and construction of the utility matrix

The analysis is based on transaction data from the online store of *Fondo de Cultura Económica Chile*. The e-commerce platform is operated through Shopify and records customer purchases made through the online channel between October 2019 and December 2025.
This setting is particularly relevant for collaborative filtering because the online store represents only one of several sales channels of the Chilean branch. In addition to the e-commerce platform, Fondo de Cultura Económica Chile operates through physical bookstores, a distribution network of client bookstores, and institutional sales. Therefore, the absence of an observed online purchase should not be interpreted as a true negative preference.

The original unit of observation corresponds to products purchased by customers within orders. For the recommendation task, the data were aggregated at the customer-item level.
Users are identified through anonymized customer identifiers. Items are defined through product identifiers based on ISBN information, with ISBN-10 codes converted into ISBN-13 format by adding the `978` prefix and recomputing the final check digit. This step reduces artificial duplication between older and newer identifiers of the same title. Shipping charges and discounts were excluded, while purchasable bookstore catalogue products were retained.
Only transactions with sufficient evidence of completed fulfillment were retained. Canceled, unpaid, returned, or unsuccessfully fulfilled orders were removed, since the objective is to model realized purchase behavior.

Each observed customer-item pair was encoded as a positive interaction. The resulting utility matrix is binary:

$$X_{ui} = \begin{cases} 1, & \text{if user } u \text{ purchased item } i, \\ 0, & \text{otherwise.} \end{cases}$$

Although the dataset preserves the number of purchased units and the number of distinct orders for each customer-item pair, the matrix factorization models use purchase occurrence as the main response variable. This choice is empirically justified: only 5.46% of observed customer-item interactions involve more than one unit, and only 1.48% correspond to repeated purchases of the same item by the same customer. Therefore, purchase incidence provides a more stable signal than purchase intensity.

3.2 Dataset summary and sparsity

After preprocessing and aggregation, the modeling dataset contains:

Quantity	Value
Users	23,605
Items	4,405
Positive user–item interactions	136,832
Matrix density	0.13%
Matrix sparsity	99.86%
Total units purchased	150,234
Interactions with more than one unit	5.46%
Repeated user–item purchases	1.48%

The matrix is therefore extremely sparse: fewer than two out of every one thousand possible user-item pairs are observed.

The high sparsity of the user-item matrix is a structural feature of the problem rather than a data defect. Fondo de Cultura Económica has a broad and heterogeneous catalogue, including social sciences, literature, humanities, children's books, and other bookstore products. Consequently, an individual customer is expected to purchase only a small fraction of the available catalogue. Moreover, because the online store is only one possible purchasing channel, some customer preferences may be expressed through physical bookstores, distributor bookstores, or institutional channels instead of the e-commerce platform. This motivates the use of collaborative filtering methods designed for sparse implicit-feedback data.

3.3 Minimum interaction thresholds

To evaluate the effect of user activity and sparsity, we constructed several filtered datasets by imposing minimum interaction thresholds τ on users.

τ	Users	Items	Interactions	Sparsity
3	11,878	4,319	121,671	99.76%
5	7,802	4,268	107,815	99.68%
10	3,665	4,101	80,536	99.46%
20	1,313	3,793	49,226	99.01%
25	916	3,684	40,601	98.80%

Increasing the threshold τ produces denser matrices, but even the strictest filtering criterion leaves the recommendation problem highly sparse.

3.4 Train-test split, negative sampling and evaluation

Split: We use a random leave-one-out split at the user level: after applying τ, one observed item was randomly held out for each user as the test interaction, while the remaining interactions were used for training. Test interactions on items that never appear during training (cold items) are removed, so that each item evaluated has a learned representation. All models share this split, with a single fixed seed.

Negative sampling: Unobserved entries are never treated as true negatives. For the two sampled-support models (NMF, Logistic MF) we build a training support

$$\Omega = \mathcal{P} \cup \mathcal{N}, \qquad |\mathcal{N}| = r\,|\mathcal{P}|,$$

where $\mathcal{P}$ is the set of training positives and $\mathcal{N}$ is a sample of r unobserved cells per positive, assigned target 0. The main protocol uses $r_{train} = 4$. For evaluation we report RMSE under two negative ratios, $r_{\text{eval}} \in \{4, 100\}$; unless stated otherwise, RMSE values quoted in the text refer to $r_{\text{eval}} = 100$. Negatives are drawn *once* and reused across the whole k-sweep, so Ω is fixed and different k are directly comparable. Cells outside Ω are left out of the objective entirely: they are unknown, not zero. iALS, by construction, does not sample, it uses all $n \cdot m$ cells.

Pointwise Metric: As the bridge to the reconstruction error of the reference paper we report RMSE on a held-out set consisting of the test positives plus r sampled negatives per positive. The metric is well defined even when the predicted scores are not constrained to [0,1], since it simply measures the squared residual $(X_{ui} - \hat{X}_{ui})^2$.

Ranking Metrics: The primary metric is NDCG@K for $K \in \{5, 10, 20\}$. For each test user, we score the entire catalogue with the raw model scores, mask the user's training positives to $-\infty$, and determine the rank of the single held-out positive. Full-catalog ranking avoids the bias that sampled-negative evaluation is known to introduce. NDCG@K (Normalized Discounted Cumulative Gain) is a ranking-aware metric designed to evaluate the quality of the recommended list. It is based on the assumption that highly relevant items are more useful to the user when they appear higher in the ranking. To compute NDCG, we first calculate the DCG (Discounted Cumulative Gain), which sums the relevance scores of the items in the top-k list, applying a logarithmic discount function based on their positions. This means that relevant items placed lower in the ranking are penalized and contribute less to the final score. Since the absolute value of DCG can vary depending on the total number of relevant items for a specific user, it is normalized by dividing it by the IDCG (Ideal DCG). The IDCG represents the maximum possible DCG

score, obtained if the system ranked the top-k items in perfect, decreasing order of relevance. Consequently, NDCG@k always yields a value between 0 and 1.

$$\text{DCG@}k = \sum_{i=1}^{k} \frac{rel_i}{\log_2(i+1)} \tag{1}$$

$$\text{NDCG@}k = \frac{\text{DCG@}k}{\text{IDCG@}k} \tag{2}$$

4 Non-Negative Matrix Factorization

Nonnegative matrix factorization (NMF) aims at decomposing a given nonnegative data matrix $\mathbf{X}$ as $\mathbf{X} \approx \mathbf{P}\mathbf{Q}^\top$ with $\mathbf{P} \geq 0$ and $\mathbf{Q} \geq 0$ (meaning that $\mathbf{P}$ and $\mathbf{Q}$ are component-wise nonnegative).
Here $\mathbf{X}$ represents the binary interaction matrix, while $\mathbf{P} \in \mathbb{R}^{n \times k}$ is the user factors matrix and $\mathbf{Q} \in \mathbb{R}^{m \times k}$ is the item factors matrix, so the prediction is $\hat{\mathbf{X}} = \mathbf{P}\mathbf{Q}^\top$. Because pure-positive training degenerates, in the sense that predicting 1 everywhere gives zero training error, we build the support Ω by adding, per user, r negative samples (unpurchased books, target 0) for every positive.
Each cell in Ω therefore carries an explicit target: 1 on observed positives and 0 on sampled negatives, while every remaining unobserved cell is left out of Ω and treated as unknown, not as zero. Negatives are sampled once and reused across the whole k-sweep, so Ω is fixed and different k values are comparable.

The NMF minimizes the masked squared reconstruction error with Tikhonov regularization:

$$\min_{\mathbf{P},\mathbf{Q} \geq 0} \sum_{(u,i) \in \Omega} \left(X_{ui} - P_u Q_i^\top \right)^2 + \lambda \|\mathbf{P}\|_F^2 + \lambda \|\mathbf{Q}\|_F^2, \tag{3}$$

where $P_u, Q_i \in \mathbb{R}^k_{\geq 0}$ are the u-th and i-th rows of $\mathbf{P}$ and $\mathbf{Q}$.

Almost all NMF algorithms designed for (3) use a two-block coordinate descent scheme: they optimize alternatively over one of the two factors while keeping the other fixed, since each subproblem is convex. In particular, we use multiplicative updates evaluated only on Ω. Let $\Omega_u = \{\, i : (u,i) \in \Omega \,\}$ be the set of items sampled for user u (its positives *and* its sampled negatives) and, symmetrically, $\Omega_i = \{\, u : (u,i) \in \Omega \,\}$ the set of users sampled for item i. The updates are

$$P_u \leftarrow P_u \odot \frac{\sum_{i \in \Omega_u} X_{ui}\, Q_i}{\sum_{i \in \Omega_u} \hat{X}_{ui}\, Q_i \;+\; \lambda P_u \;+\; \varepsilon},$$

$$Q_i \leftarrow Q_i \odot \frac{\sum_{u \in \Omega_i} X_{ui}\, P_u}{\sum_{u \in \Omega_i} \hat{X}_{ui}\, P_u \;+\; \lambda Q_i \;+\; \varepsilon},$$

where $\varepsilon = 10^{-9}$ guards against a vanishing denominator. Note that the regularization term appears *outside* the sum: the gradient of $\lambda \|P_u\|_F^2$ does not depend on how many cells of Ω user u contributes.

The numerator vanishes on sampled negatives ($X_{ui} = 0$), whereas the denominator runs over all cells of Ω, so negatives still shape the update through $\hat{X}$. With a uniform $[0, 1]$ initialization these updates keep $\mathbf{P}$ and $\mathbf{Q}$ nonnegative at every step.

Both the objective and the updates are evaluated only on the cells in Ω (observed positives plus sampled negatives), never on the full $n \times m$ matrix.

4.1 Experiments and Results

The goal of this section is not merely to check whether an overparameterized NMF attains a low error, but to test under which conditions the double descent phenomenon reported by Kawakami and Sugiyama survives when it is transferred from dense movie-rating benchmarks to a real, sparse, one-class e-commerce matrix. For this purpose several experiments have been conducted to study the effect of filtering the users with higher activity, of using different metrics and of decreasing the regularization term. Let $\tau \in \{3, 5, 10, 20\}$ be the minimum number of interactions a user must have to be kept in the analysis.
In the first experiment we set $\tau = 3$, $\lambda = 0.06$ and, to overcome the computational cost, different numbers of epochs:

- $k \leq 512$: 100 epochs,

- $k \leq 2048$: 60 epochs,

- $k = 3072$: 40 epochs.

At k = 1, RMSE rises from 0.247 (with negative sampled ratio 1:100) to 0.349 (with negative sampled ratio 1:4): with fewer negatives the model doesn't look so good, confirming that its low 1:100 RMSE is partly the negative-heavy sampling artifact. Importantly, both ratios still rise to the peak and fall afterward (no hidden first descent emerges) so the shape is a genuine property of the setting, not merely an artifact.

The train RMSE decreases smoothly from 0.335 and plateaus around 0.03, never approaching zero, because $\lambda = 0.06$ prevents the factors from growing enough to null the residuals. At the test peak ($k \approx 24$) the train error is still 0.20. Therefore the test peak does not coincide with literal interpolation of Ω. The peak is better read as an overfitting peak: at $k \approx 24$ the model gains just enough capacity to personalize beyond a popularity baseline, but with very little signal per user it fits noise, followed by the implicit-regularization benefit of the overparameterized regime. In fact the second descent is attributable to the algorithm's implicit inductive bias: in NMF multiplicative updates are not standard gradient descent and the structural bias is non-negativity, for this reasons there is a preference for low effective rank of the reconstructed matrix (differently from the regression case).

On the other hand, the NDCG metrics show the classical double descent trend,

although in reverse, since higher values indicate improved performance. As showed in figure 2 the curve reaches the first peak at k=2, then drops to a minimum at k=32, and subsequently rises as the number of latent factors increases, eventually surpassing the performance of the first peak.

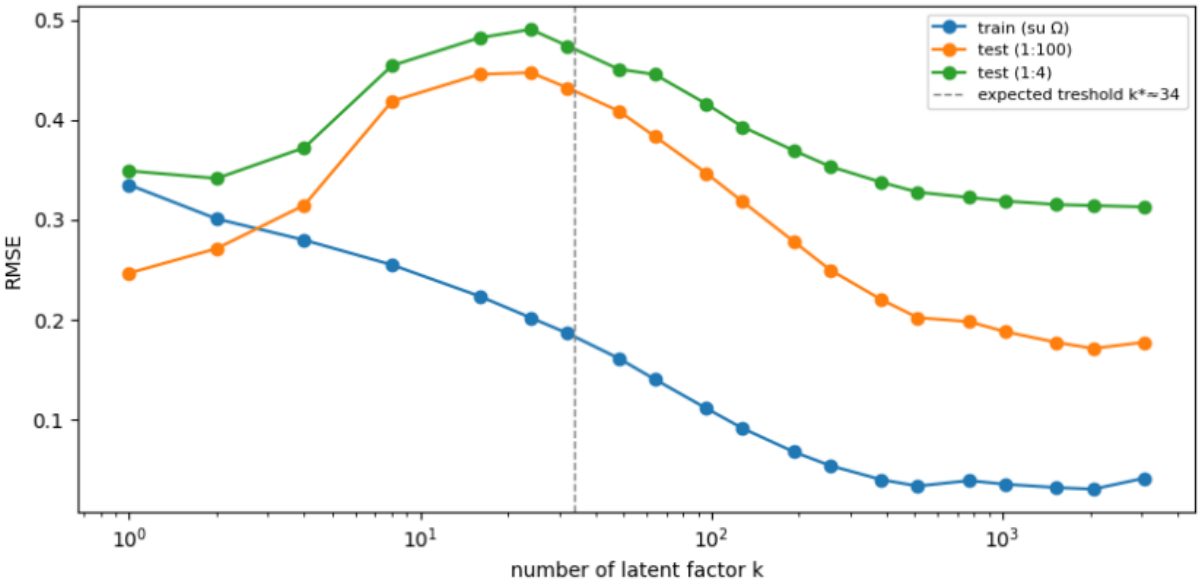


Figure 1: Train and Test RMSE for $\tau = 3$ and $\lambda = 0.06$

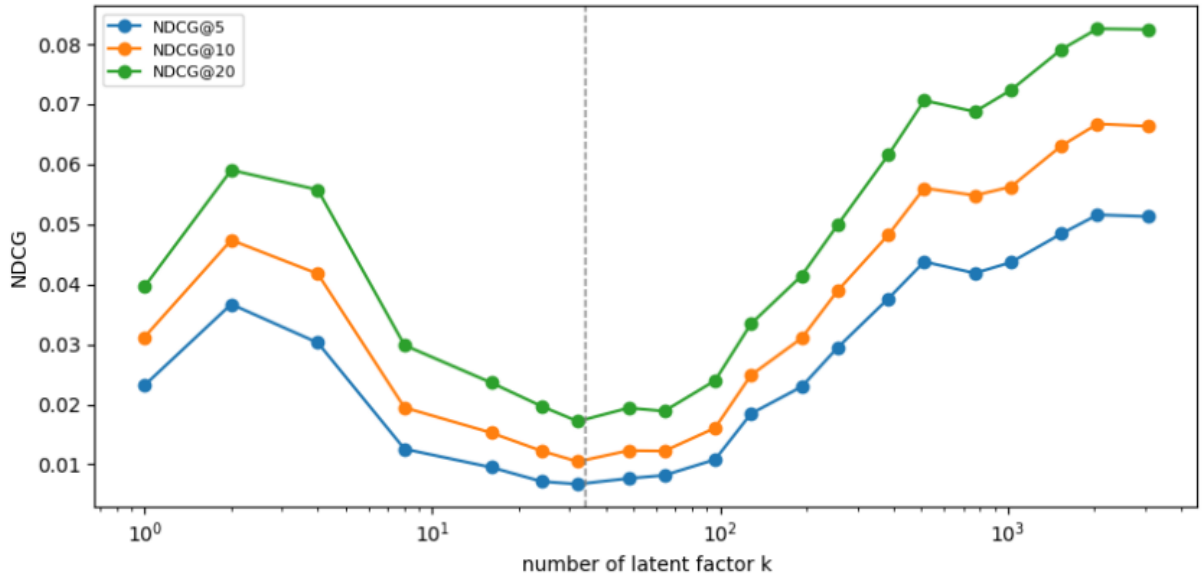


Figure 2: NDCG for $\tau = 3$ and $\lambda = 0.06$

Since for the RMSE the curve rises from k=1 to the peak rather than descending, so the classical underparameterized regime is missing, our main hypothesis is the sparse one-class signal: with so few positives per user a low capacity model already

captures most of the explainable signal. There is no "under-capacity" regime to descend from; additional factors mostly overfit noise. This hypothesis is testable by raising the filter threshold τ: richer users have profiles that 1–2 factors cannot capture, so a first descent should appear if the hypothesis holds.

In the next experiments the same setting is used (same masked NMF, same multiplicative updates, $\lambda = 0.06$, raw scores for ranking, full-catalogue ranking) under three progressively stricter activity filters, $\tau \in \{5, 10, 20\}$.
Across all three filters the global structure is preserved: a smoothly decreasing train error, a pronounced test-error peak in the critical region, and a strong second descent into the overparameterized regime. The new feature, absent (or barely present) at $\tau = 3$ and $\tau = 5$, is a small-k regime in which increasing k improves test performance before the critical worsening, i.e. the first descent. So the central result is that the first descent is not a fixed property of the data but a function of per-user density. The mechanism is the one anticipated: when most users have only few activities, a k = 1 model already captures nearly all the explainable signal, so there is no under-capacity regime to descend from; the curve starts already at (or near) its critical rise. As we retain users with $\tau = 10$ and $\tau = 20$ interactions, their profiles acquire a structure that a k=1 or k=2 models cannot represent, so a few additional factors now are useful. The classical underparameterized regime is thus restored by enriching the signal, not by changing the model.

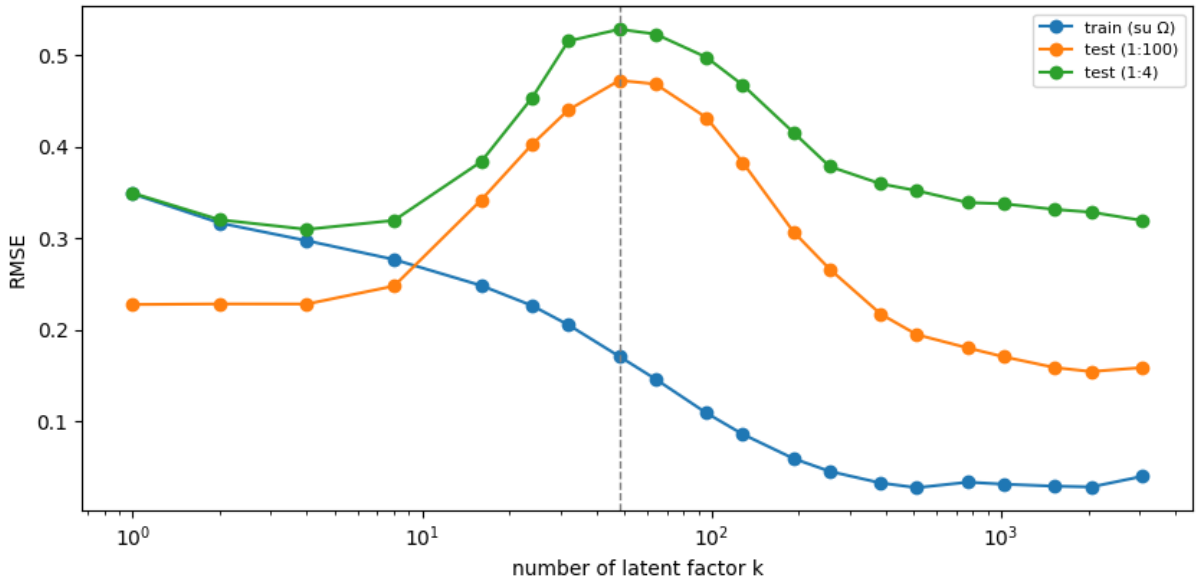


Figure 3: Train and Test RMSE for $\tau = 20$ and $\lambda = 0.06$

NDCG in figure 4 displays the double descent more sharply than the RMSE, because it measures the ranking order, while RMSE measures the absolute value alignment. At $k = 1$, the model can only generate a uniform popularity baseline for everyone; personalization (user-specific ranking) only begins at $k \approx 2 - 4$. Because NDCG is purely ordinal and immune to score scale, it immediately captures this first personalization shift as a sharp performance rise across all filters (τ). Conversely, RMSE is computed on an evaluation set dominated by sampled negatives, which dilutes the contribution of the single held-out positive. A pure ranking improvement barely registers in RMSE and is often masked by noisy negative score shifts. Thus, RMSE requires dense user activity ($\tau = 20$) to accumulate a signal strong enough to overcome this dilution and finally show a marked double descent.

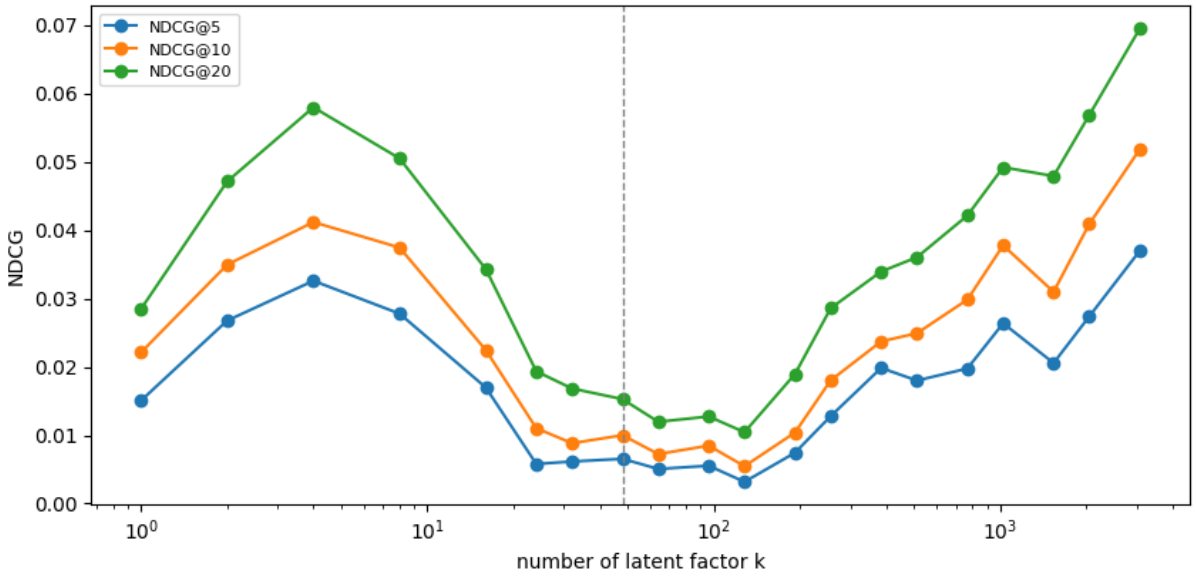


Figure 4: NDCG for $\tau = 20$ and $\lambda = 0.06$

Restoring the first descent does not affect the second one. In every filter, the heavily overparameterized model remains the best on all metrics: the lowest test RMSE and the highest NDCG@K are attained at the largest k. The ≥ 10, and in particular the ≥ 20 filter, as shown in figure 3 display the classic double descent shape that the ≥ 3 only partially exhibited.
Repeating the $\tau = 20$ sweep with a weaker Tikhonov penalty ($\lambda = 0.01$) confirms that weaker regularization amplifies the critical peak.

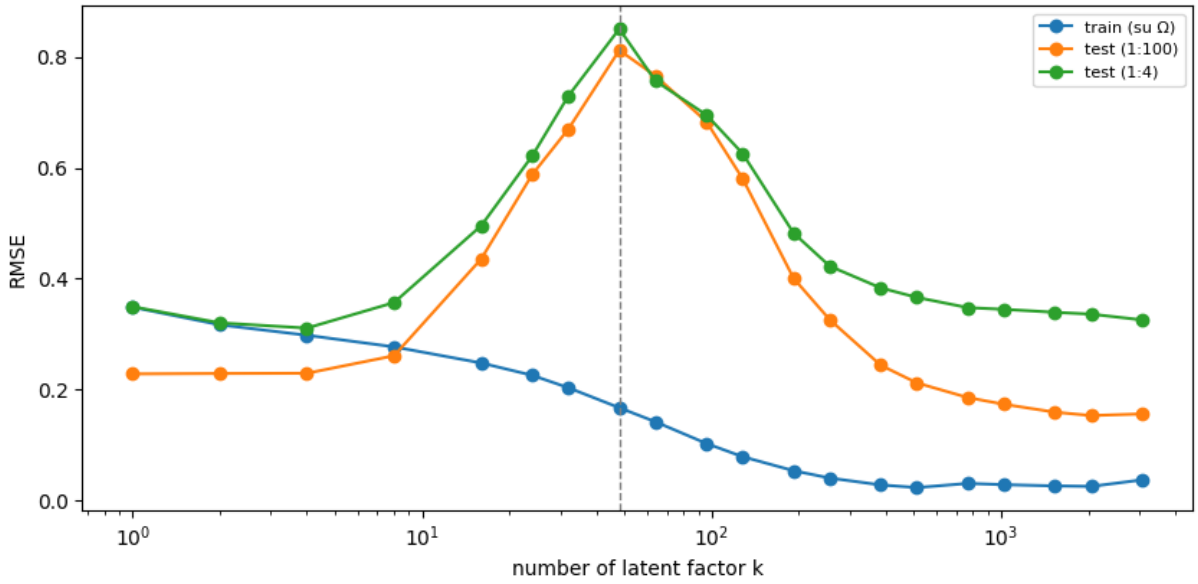


Figure 5: Train and Test RMSE for $\tau = 20$ and $\lambda = 0.01$

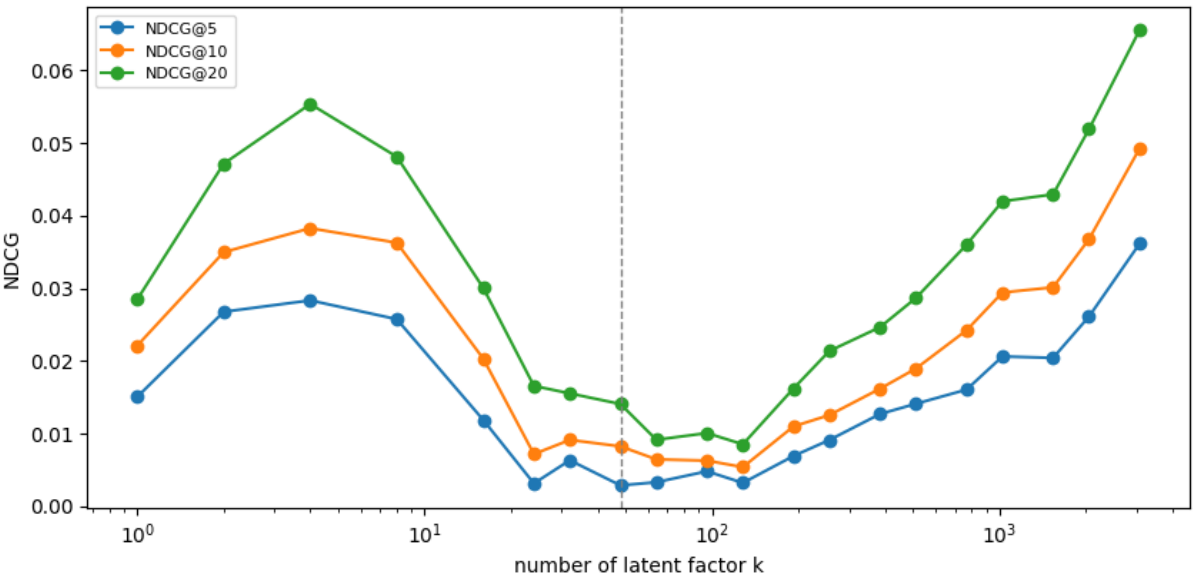


Figure 6: NDCG for $\tau = 20$ and $\lambda = 0.01$

The peak of the test RMSE (1:100) rises from ≈ 0.47 at $\lambda = 0.06$ to ≈ 0.82 at $\lambda = 0.01$, therefore the double descent curve is more pronounced under light regularization, exactly as observed in the original paper. Regularization does not create or destroy the phenomenon: it modulates its amplitude. Furthermore, the metric asymmetry is invariant to λ: the NDCG curves retain the identical shape they had at $\lambda = 0.06$, while the RMSE only barely registers the small-k gain.

4.2 Discussion

The NMF experiments lead to a clear but conditional interpretation. The sweep over k shows a pronounced test peak in the critical region followed by a strong second descent, at every activity filter and on both metrics. What changes with the filter is not the second descent but the first: at $\tau = 3$ the curve begins already at its critical rise, because with one or two positives per user a $k = 1$ model already captures nearly all the explainable signal and there is no under-capacity regime to descend from. At $\tau = 10$ and $\tau = 20$ the classical descent reappears. This indicates that the underparameterized regime is a property of the per-user density of the data, not of the model. The two metrics do not resolve the phenomenon equally well and Weakening λ from 0.06 to 0.01 raises the peak from ≈ 0.47 to ≈ 0.82 without moving its location and without altering the NDCG curve, confirming that regularization modulates the amplitude of the phenomenon rather than creating or destroying it.

5 Weighted Matrix Factorization (Implicit ALS)

Weighted Matrix Factorization, commonly known as implicit Alternating Least Squares (iALS), follows the formulation of Hu, Koren and Volinsky (2008) for implicit-feedback, one-class data. Unlike NMF with negative sampling, iALS does not restrict training to a sampled support Ω: every cell of the $n \times m$ user–item matrix is used at every iteration, positives and non-observed cells alike, and the two are distinguished only through a per-cell *confidence* weight rather than through sampling.

Let $X_{ui} \in \{0, 1\}$ be the binary preference (1 if user u purchased item i, 0 otherwise) and let $C_{ui} = 1 + \alpha X_{ui}$ be the associated confidence, where α controls how much more an observed purchase should be trusted relative to an unobserved cell. With $P \in \mathbb{R}^{n \times k}$ and $Q \in \mathbb{R}^{m \times k}$ the user and item factor matrices, the model minimizes

$$\min_{P,Q} \sum_{u=1}^{n} \sum_{i=1}^{m} C_{ui} \Big(X_{ui} - P_u^\top Q_i \Big)^2 \; + \; \lambda \Big(\|P\|_F^2 + \|Q\|_F^2 \Big), \tag{4}$$

where $p_u, q_i \in \mathbb{R}^k$ are the u-th and i-th rows of P and Q. Because the objective is quadratic and convex in each factor when the other is fixed, it is optimized with alternating least squares: fixing Q, each user row has the closed-form solution

$$p_u = \Big(Q^\top C^u Q + \lambda I \Big)^{-1} Q^\top C^u X_u, \tag{5}$$

and symmetrically for q_i with P fixed, where $C^u = \mathrm{diag}(C_{u1}, \ldots, C_{um})$ and x_u is the u-th row of X. In practice the $m \times m$ (resp. $n \times n$) linear systems are never formed explicitly: since $C_{ui} = 1$ for all non-observed cells, Equation 5 can be rewritten so that the cost of one update scales with the number of *observed* interactions rather than with $n \cdot m$, and the underlying linear system is solved with a few steps of conjugate gradient. We use the `implicit` Python library (v0.7.3), which implements exactly this scheme; we verified directly in its source that, for binary data, the library computes $C_{ui} = 1 + \alpha X_{ui}$, matching Equation 4.

With uniform random initialization of P and Q, the two half-updates are repeated for a fixed number of iterations. Unlike NMF, no non-negativity constraint is imposed on P, Q, and no explicit negatives are sampled: the entire matrix acts as training signal at every step, with unobserved cells contributing through $C_{ui} = 1$ rather than being ignored.

5.1 Experiments

A single sweep over k would be weak evidence on its own, regardless of the model: the shape of the curve could be an artefact of an insufficiently trained model rather than a genuine effect of capacity. Before trusting any result at large k, we therefore had to establish how many ALS iterations are actually needed for the model to converge, since this is not fixed a priori.

5.1.1 Convergence check

We compared iterations $\in \{15, 30, 60, 100\}$ at two representative values of k. At k=128 (near the peak of the curve), NDCG@20 improves with more iterations but with rapidly shrinking increments (+1.6%, +0.9%, +0.2% relative to the previous iteration count), the signature of genuine convergence. At k=3072 (heavily overparameterized relative to n and m), the opposite happens: NDCG@20 *worsens monotonically* with more training (−27.6%, −43.9%, −54.5% relative to iterations= 15), with no sign of stabilizing even at 100 iterations. This is not under-convergence in the usual sense — it is training-time overfitting for very large k, analogous to the epoch-wise double descent reported in the deep learning literature, but along the training-time axis rather than the capacity axis. We verified directly (via a per-iteration snapshot of $\|P\|_F$) that at k=3072 the factors keep changing across iterations, ruling out a trivial bug. Given this, we fixed **iterations**= 50 for every k in the definitive sweep: close to full convergence for small/medium k, without letting the largest k overfit further than necessary. All results reported below are computed on CPU, for every k and every user-activity threshold.

5.1.2 Results

Main sweep, standard threshold ($\tau = 3$): Figure 7 shows RMSE and NDCG@$\{5, 10, 20\}$ against $k \in [1, 3072]$, iterations= 50. The curve has a single, well-defined peak at k=128 (NDCG@20= 0.212) and declines monotonically for larger k, down to NDCG@20= 0.032 at k=3072. There is no second improvement anywhere in the tested range: on this population, we find *no evidence of double descent* for Weighted MF / iALS. The gap between the iterations = 50 run and the historical iterations = 15 run widens with k: iterations= 50 is slightly *better* for $k \leq 512$ and markedly *worse* for $k \geq 768$, consistent with the training-time overfitting described above.

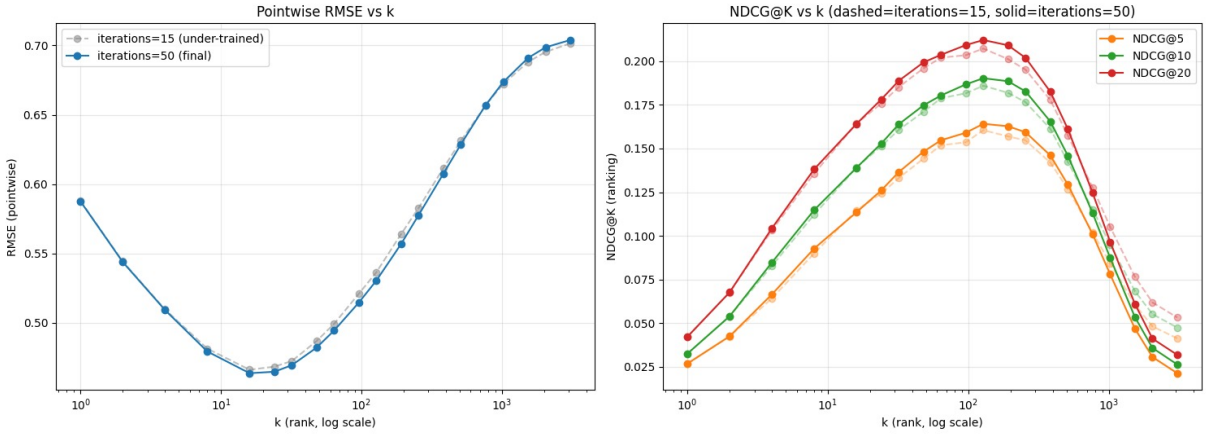


Figure 7: iALS, $\tau = 3$: RMSE (left) and NDCG@K (right) vs. k, iterations= 50 (solid) compared to the historical iterations= 15 run (dashed).

Threshold $\tau = 5$. The same sweep repeated on a smaller, slightly denser population (Figure 8) reproduces the same qualitative shape: a single peak, now at k=128 (NDCG@20= 0.182, over a broad plateau spanning $k \in [96, 192]$), followed by an uninterrupted decline to NDCG@20= 0.027 at k=3072. Again, no recovery.

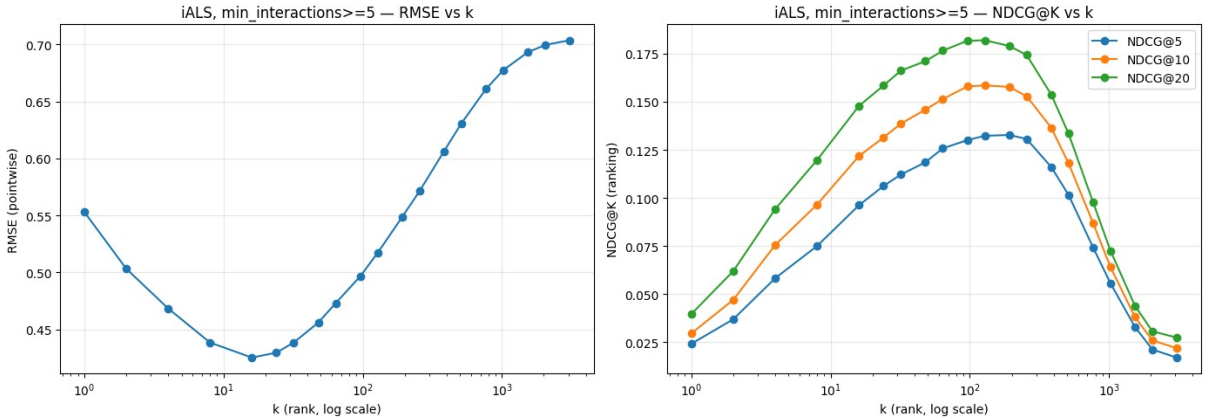


Figure 8: iALS, $\tau = 5$: RMSE (left) and NDCG@K (right) vs. k, iterations= 50.

Threshold $\tau = 10$. With a smaller population (Figure 9) the curve still peaks early (NDCG@20= 0.142, $k = 96$) and declines for most of the range, reaching a minimum at k=2048 (NDCG@20= 0.028) — but, unlike $\tau = 3$ and $\tau = 5$, the last point of the grid shows a genuine uptick: NDCG@20 rises to 0.037 at k=3072, a +34.7% increase over the minimum. This recovery is real (verified numerically) but weak — only 26% of the original peak is recovered by the end of the tested range, and since $k = 3072$ is our largest grid point we cannot confirm whether the curve keeps rising beyond it.

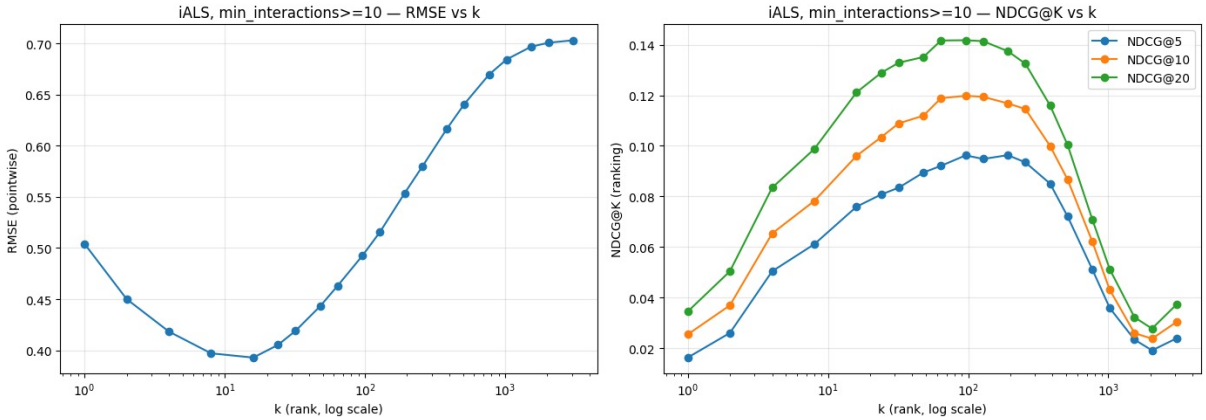


Figure 9: iALS, $\tau = 10$: RMSE (left) and NDCG@K (right) vs. k, iterations= 50.

Threshold $\tau = 20$. This population makes the recovery unambiguous (Figure 10). NDCG@20 peaks at k=128 (0.106), falls to a minimum at k=768 (0.042), and then rises *monotonically* across every remaining grid point (k=1024, 1536, 2048, 3072) up to 0.079 — 74.5% of the original peak. Unlike $\tau = 10$, this is not a single uptick at the edge of the grid but a clean, sustained recovery spanning four consecutive points, which makes it much stronger evidence of a genuine double-descent shape.

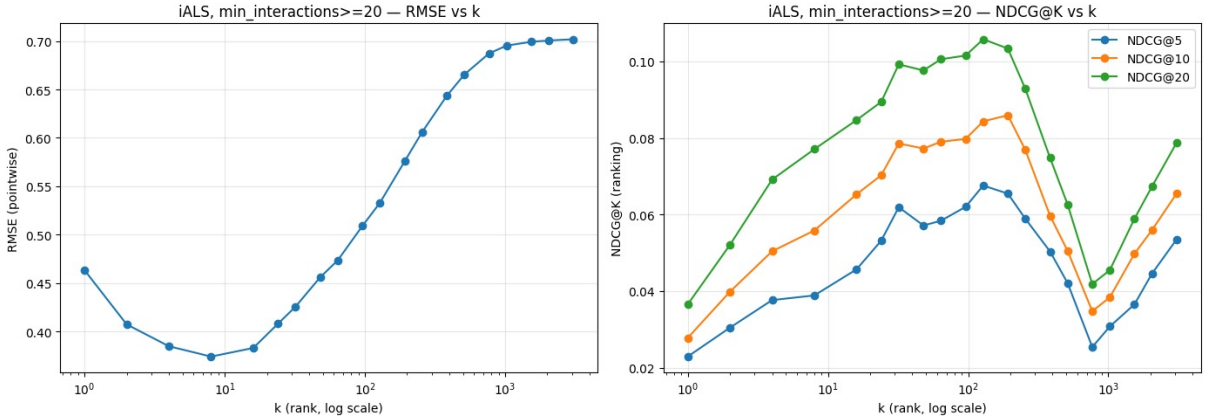


Figure 10: iALS, $\tau = 20$: RMSE (left) and NDCG@K (right) vs. k, iterations= 50.

Threshold $\tau = 25$. This is the smallest and densest population tested. At this population, the iALS curve changes qualitatively (Figure 11): NDCG@20 rises to a first local peak at k=24 (0.093), falls to a minimum at k=768 (0.044), and then *recovers* to 0.081 at k=3072 — 87.5% of the original peak. This is the strongest and most complete double-descent recovery observed across all five thresholds. Notably, the recovery appears only in the ranking metric: RMSE at the same $\tau = 25$ rises essentially monotonically from 0.449 (k=1) to 0.702 (k=3072), with

no second descent — the double-descent signature is visible in NDCG but not in the pointwise error, for this model and this population.

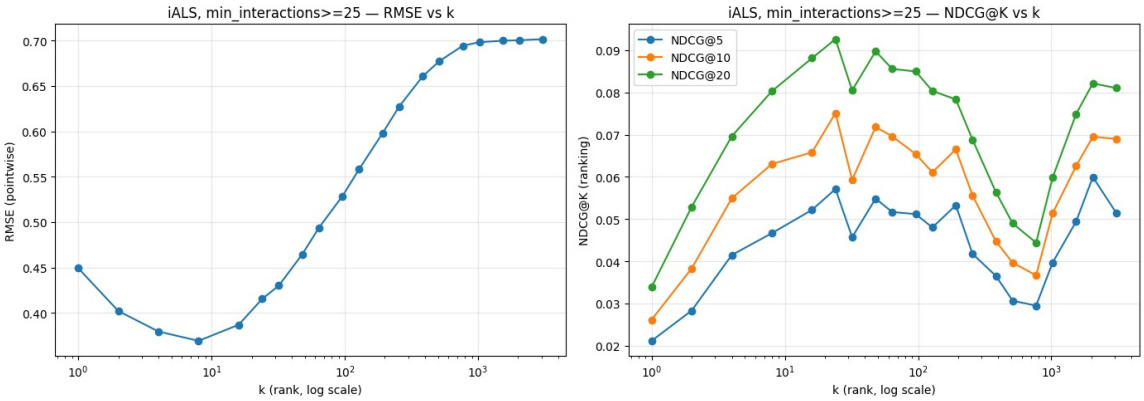


Figure 11: iALS, $\tau = 25$: RMSE (left) and NDCG@K (right) vs. k, iterations= 50.

Having swept five thresholds ($\tau = 3, 5, 10, 20, 25$) with the same methodology, a clearer picture emerges than a simple presence/absence of double descent. For each τ, Table 1 reports the NDCG@20 peak, its location, the value at the last grid point (k=3072), and the fraction of the peak retained at k=3072 — a single number that captures how much of the initial gain survives (or is recovered) at extreme overparameterization, regardless of whether the curve dips and rises again or simply decays.

τ	Peak NDCG@20	NDCG@20 at k=3072	Fraction of peak
3	0.212 (k=128)	0.032	15.1%
5	0.182 (k=128)	0.027	15.0%
10	0.142 (k=96)	0.037	26.3%
20	0.106 (k=128)	0.079	74.5%
25	0.093 (k=24)	0.081	87.5%

Table 1: NDCG@20 retained at k=3072 as a fraction of each curve's own peak, across the five user-activity thresholds tested.

Table 1 uses the same $k_{\max} = 3072$ for every threshold, but $n + m$ shrinks as τ grows, so $k_{\max}$ reaches a very different *relative* depth of overparameterization at each τ: only $0.19\times(n+m)$ at $\tau = 3$, versus $0.67\times(n+m)$ at $\tau = 25$ (Table 2, second column). This means part of the apparent gradient could simply reflect that we probed further into the overparameterized regime, in relative terms, for the smaller populations — not a genuine effect of population size or density. To check this, we re-examined each curve only up to the k that matches $\tau = 3$'s relative depth

$(0.19 \times (n+m))$, i.e. the same "distance travelled" for every threshold.

τ	$k_{\max} = 3072$ depth	Matched k	NDCG@20 at matched k	Peak	Fraction
3	0.19×	3072	0.032	0.212	15.1%
5	0.26×	2048	0.031	0.182	16.8%
10	0.40×	1024	0.051	0.142	36.1%
20	0.60×	768	0.042	0.106	39.5%
25	0.67×	768	0.044	0.093	47.9%

Table 2: Same five curves, each truncated to the largest k in its own grid that does not exceed $\tau = 3$'s relative depth ($k/(n+m) \leq 0.19$).

Two things follow from this. First, the scale confound is real and inflates the size of the effect reported in Table 1: roughly half of the apparent gradient (15% → 87%) is attributable to smaller populations simply being probed further into the over-parameterized regime with the same $k_{\max}$. Second, the confound does not fully explain it: at matched relative depth, $\tau = 10/20/25$ still retain markedly more of their peak (36–48%) than $\tau = 3/5$ (15–17%, statistically indistinguishable from each other). We therefore report the effect as a genuine but more modest gradient than our first pass suggested, and we do not claim to have isolated a pure population/density effect from a pure capacity-depth effect — disentangling the two rigorously would require extending K_{GRID} for $\tau = 3$ and $\tau = 5$ to the same relative depth reached at $\tau = 25$ (i.e. $k \approx 10{,}800$ and $k \approx 8{,}000$ respectively), which we did not have time to run given the CPU cost already observed at $k = 3072$ (already 261.6s to fit a single model at $\tau = 3$; a further 3–4× increase in k would be considerably slower still).

5.2 Discussion

Double descent in iALS is not simply absent: at matched relative depth, the smaller/denser populations ($\tau \geq 10$) retain roughly twice the fraction of their peak that the two larger populations do, so *some* genuine population-dependent effect survives the scale correction, even if we cannot yet pin down its full size. A speculative explanation is that iALS fits the *entire* $n \times m$ matrix with an explicit ℓ_2 penalty at every k, so there is no finite, fixed set of cells it can exactly interpolate the way a sampled-support model could — which would make the double-descent mechanism weaker and more dependent on how sparse or dense the population is. This is a hypothesis consistent with what we observed, not something we verified directly.
A few things were not fully controlled for and should be stated plainly. The

iterations= 50 choice was validated (via the convergence check) only at $\tau = 3$, at $k = 128$ and $k = 3072$; we did not re-verify that 50 iterations is neither too few nor too many at $\tau = 10, 20, 25$, where $k = 3072$ is an even more extreme multiple of n, and the training-time overfitting documented at $\tau = 3$ could in principle also be affecting the smaller populations. Every run reported here uses a single fixed seed for both the data split and the model's random initialization; the weaker signals, in particular the single-point uptick at $\tau = 10$, have not been checked for robustness across multiple random seeds and could partly reflect the variance of one particular run. Finally, this five-threshold sweep was exploratory: we did not fix $\tau = 3, 5, 10, 20, 25$ in advance as a pre-registered design, but added intermediate thresholds after $\tau = 25$ showed an unexpected recovery, specifically to test whether the effect was a gradient or an isolated case. The resulting pattern should be read as hypothesis-generating evidence, not as a confirmatory test of a pre-specified claim.

6 Logistic Matrix Factorization Model

Instead of approximating the user-item matrix directly like NMF and WMF, Logistic MF models the probability that a user interacts with an item. This is especially appropriate for binary purchase data, where the main signal is whether a customer bought an item at least once.

For each user u and item i, Logistic Matrix Factorization defines the interaction probability as

$$P(y_{ui} = 1) = \sigma\big(\mu + b_u + b_i + p_u^\top q_i\big)\,,$$

where p_u and q_i are user and item latent vectors, b_u and b_i are user and item bias terms, μ is a global intercept, and $\sigma(\cdot)$ is the logistic function.

The model is trained by minimizing binary cross-entropy over a sampled support. Observed purchases are treated as positive examples, while unobserved user-item pairs are sampled as negative examples. In the main protocol, four negative items are sampled for each positive interaction. These sampled negatives are not interpreted as explicit dislikes. Instead, they provide contrastive observations that allow the model to distinguish observed purchases from plausible unobserved alternatives.

Evaluation follows the same leave-one-out protocol used in the rest of the project. For each user, one observed interaction is held out as the test item, and training positives are masked during recommendation. Pointwise performance is measured using RMSE with sampled negatives, including a test set with 100 negatives per positive interaction. Recommendation quality is evaluated using NDCG@K.

Unless otherwise stated, comparisons use the same data split, negative sampling protocol, regularization strength, evaluation samples, and fixed random seed. Models with different values of k or different training budgets are trained as independent deterministic runs.

6.1 Experiments and Results

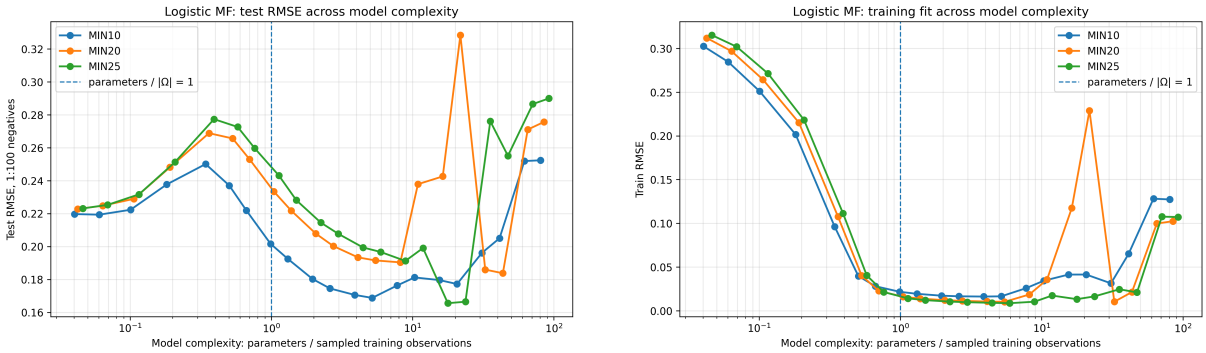
6.1.1 Initial 30-epoch capacity sweep

We first trained Logistic MF over a broad range of latent dimensions using 30 epochs and a learning rate of 0.01. The resulting test-error curve showed the qualitative structure initially sought in the project. Test RMSE worsened in an intermediate-capacity region and subsequently improved as the latent dimension increased, producing a curve visually compatible with a double-descent-like recov-

ery.

However, the initial experiment also showed an additional and unexpected increase at the largest latent dimension. This second high-capacity peak could not immediately be interpreted as another generalization phenomenon. At the same value of k, training error also increased sharply.

This observation is important for interpretation. Under classical overfitting, we would expect training error to remain low while test error increases. A simultaneous deterioration in training and test error instead suggests that the model has not been successfully optimized. Therefore, the initial sweep produced two distinct observations: a candidate double-descent-like recovery over the main capacity range and an additional high-k instability that required further investigation.



(a) Test RMSE (b) Train RMSE

Figure 12: Initial 30-epoch Logistic MF sweep with $lr = 0.01$. Test RMSE displays a double-descent-like recovery over the main capacity range, followed by an additional high-capacity peak. The simultaneous increase in training error at the largest model motivates an optimization diagnostic.

6.1.2 Effect of the training budget

The first diagnostic examined whether the second high-capacity peak resulted from an insufficient training budget. We retained the learning rate $lr = 0.01$ and evaluated models trained for 30, 50, 100, and 200 epochs on the $\tau = 10$ dataset. The high-capacity grid was

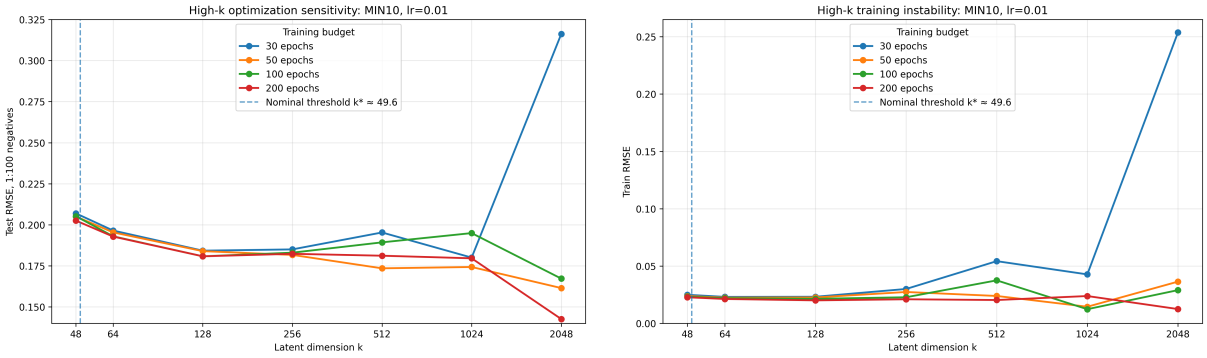
$$k \in \{48, 64, 128, 256, 512, 1024, 2048\}.$$

The data split, sampled negatives, regularization, batch size, evaluation protocol, and random seed were held fixed. The vertical line in Figure 13 indicates the nominal parameter-count threshold $k^\star \approx 49.6$, where the number of model parameters is approximately equal to the size of the sampled training support. This threshold

is used as a capacity reference and should not be interpreted as direct proof that exact empirical interpolation occurs at that value.

The 30-epoch model exhibited a sharp increase in both training and test RMSE at $k = 2048$. Training RMSE increased to approximately 0.254, while test RMSE increased to approximately 0.316. In contrast, the same catastrophic behavior was not reproduced with larger training budgets.

At $k = 2048$, test RMSE was approximately 0.161, 0.167, and 0.143 for 50, 100, and 200 epochs, respectively. Training RMSE also returned to substantially lower values. The results are not monotonic in the number of epochs, and therefore they do not imply that additional epochs always improve generalization. Nevertheless, they show that the extreme second peak is specific to the short-training configuration and is not a stable property of the model-capacity curve.



(a) Test RMSE (b) Train RMSE

Figure 13: Effect of the training budget in the high-capacity region with $lr = 0.01$. The 30-epoch run exhibits a simultaneous train and test RMSE spike at $k = 2048$. The extreme deterioration is not reproduced with 50, 100, or 200 epochs, supporting an incomplete or unstable optimization interpretation.

Some individual curves remain non-monotonic. In particular, the 100-epoch test curve decreases, rises again at intermediate-to-high dimensions, and then decreases at $k = 2048$. Although this curve is visually double-descent-like, its precise shape is not reproduced across every training budget. Consequently, the high-k experiment is interpreted primarily as an optimization-sensitivity diagnostic rather than as the main evidence for double descent.

6.1.3 Learning-rate stabilization

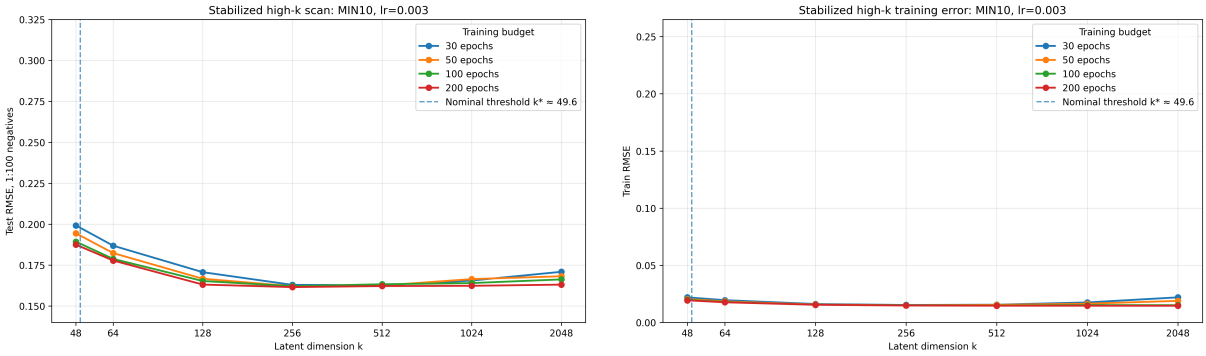
The second diagnostic investigated whether the learning rate contributed directly to the high-capacity instability. We repeated the same high-k experiment after reducing the learning rate from 0.01 to 0.003. The latent-dimension grid, training

budgets, data split, negative-sampling protocol, batch size, regularization, evaluation samples, and random seed were kept fixed.

The controlled 30-epoch comparison is particularly informative. With $lr = 0.01$, the model at $k = 2048$ produced training and test RMSE values of approximately 0.254 and 0.316, respectively. With the same seed and the same 30-epoch training budget, but with $lr = 0.003$, training RMSE fell to approximately 0.022 and test RMSE fell to approximately 0.171.

Thus, the extreme high-capacity peak disappeared without increasing the number of epochs. This result supports the interpretation that the larger learning rate produced unstable optimization in the most highly overparameterized model.

The stabilized training curves remain low across the full high-k grid for all four training budgets. Test RMSE decreases substantially between $k = 48$ and approximately $k = 256$–512. Beyond this region, it reaches a broad plateau or increases only moderately. None of the stabilized curves reproduces the catastrophic high-k peak observed under the original 30-epoch configuration.



(a) Test RMSE (b) Train RMSE

Figure 14: Learning-rate stabilization in the high-capacity region. Reducing the learning rate to 0.003 keeps training error low for every latent dimension and training budget. Test RMSE reaches a broad minimum around $k = 256$–512 and remains stable at larger dimensions, without reproducing the extreme second peak.

The training-budget and learning-rate diagnostics provide complementary evidence. Increasing the number of epochs gives the optimizer additional updates, while reducing the learning rate controls the magnitude and stability of those updates. Since both interventions remove the extreme peak, the original behavior is best explained by the interaction between large model capacity, an aggressive learning rate, and an insufficient training budget.

6.1.4 Final stabilized capacity sweep

After identifying a more stable optimization configuration, we repeated the complete Logistic MF capacity sweep. This final experiment used a learning rate of 0.003, 50 training epochs, a fixed batch size of 1024, weight decay of 10^{-6}, and gradient clipping with norm 1.0. The model specification, data split, negative-sampling ratio, and evaluation protocol remained unchanged.

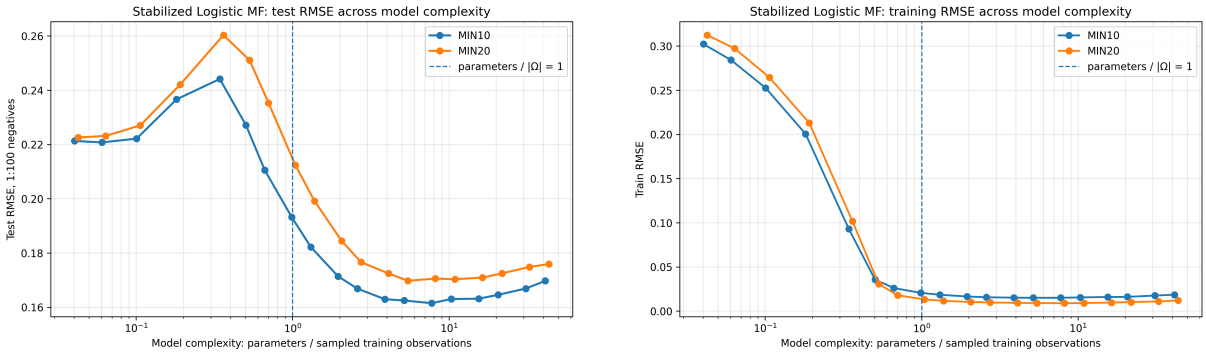
Unlike the previous high-k diagnostics, this experiment covers the full capacity range, including the low-dimensional and intermediate-dimensional regions. It is therefore the appropriate experiment for evaluating the complete shape of the generalization curve.

The clearest result is obtained under the $\tau = 20$ condition. Test RMSE initially increases as the latent dimension grows and reaches an intermediate peak around $k = 16$. It then decreases sharply as the model enters the overparameterized regime, reaching its best region around $k = 256$. For larger values, from approximately $k = 512$ to $k = 2048$, test RMSE remains on a relatively stable plateau with only mild degradation.

This sequence is consistent with a double-descent-like pattern:

$$\text{initial deterioration} \longrightarrow \text{intermediate peak} \longrightarrow \text{overparameterized recovery}.$$

The corresponding training-error curve remains well behaved and does not show the extreme high-capacity explosion observed in the original configuration. Therefore, the recovery in test performance occurs under an optimization protocol that successfully fits the sampled training support.



(a) Test RMSE (b) Train RMSE

Figure 15: Final stabilized Logistic MF capacity sweep. Under MIN20, test RMSE increases toward an intermediate-capacity peak and subsequently decreases as the model becomes overparameterized. Performance reaches its best region around $k = 256$ and then remains approximately stable at larger latent dimensions.

Ranking metrics provide complementary evidence. NDCG@K is noisier than pointwise RMSE, as expected in a leave-one-out implicit-feedback setting. Nevertheless, it improves after the low-capacity region and remain competitive for the overparameterized models. The second-descent pattern is therefore clearest in test RMSE, while the ranking metric indicates that the larger models also recover useful recommendation structures.

6.2 Discussion

The Logistic MF analysis separates two empirically different phenomena.

First, the complete stabilized capacity sweep presents a non-monotonic generalization curve consistent with double-descent-like behavior. Under $\tau = 20$, test RMSE worsens at intermediate capacity, improves after the model enters the overparameterized regime, and remains approximately stable for larger latent dimensions.

Second, the additional extreme peak observed at $k = 2048$ in the original 30-epoch experiment is not a robust generalization effect. It is accompanied by a sharp increase in training error, disappears when the training budget is increased, and also disappears when the learning rate is reduced while the 30-epoch budget is held fixed. These results support an optimization-based interpretation of the second peak.

The intermediate-capacity peak and the later high-capacity instability should therefore not be conflated. The former persists in the complete stabilized sweep and is consistent with a double-descent-like recovery. The latter depends on an unstable optimization configuration and should be interpreted as an artifact of the training procedure.

Overall, Logistic Matrix Factorization provides evidence consistent with double-descent-like generalization in this implicit-feedback recommendation task. However, the experiment also shows that the apparent shape of the capacity curve can be strongly affected by the optimization protocol. Consequently, claims of double descent should be based on configurations for which training behavior is stable and should be supported by sensitivity analyses over learning rates and training budgets.

7 Conclusion

We set out to test whether the rank-wise double descent that Kawakami and Sugiyama demonstrated for overparameterized NMF on explicit ratings, measured in reconstruction, survives when the feedback becomes one-class and implicit, and performance must be judged by ranking. Across three models and five activity filters, the answer is a qualified yes, and the qualifications are the interesting part.

Where the critical region falls is not accidental. Simple parameter counting predicts it: with $k(n+m)$ parameters against $|\Omega|$ constraints, the peak is expected near $k^\star \approx |\Omega|/(n+m)$, and this locates it correctly once Ω is fixed by negative sampling. For NMF the observed test peak sits at $k \approx 24$ against $k^\star = 34$ at $\tau = 3$, and coincides with $k^\star = 47$ at $\tau = 20$; the NDCG trough at $\tau = 3$ falls at $k = 32$, essentially on the prediction; for Logistic MF the peak falls at or slightly before normalized complexity 1. The systematic pattern is that the observed peak sits at or below $k^\star$, and approaches it as per-user density grows, because parameter counting gives an upper bound that a model can only saturate when each row carries enough signal to use its capacity. Since $k^\star$ is nearly constant across τ - raising the filter removes users and their positives in roughly equal proportion - this is a genuine prediction, not a rescaling of the axis.

The phenomenon is real but its *visibility* depends on the metric, and the direction of the asymmetry depends on the model. For NMF and iALS, NDCG is the more expressive metric: it is invariant to monotone per-row rescaling of the scores, so it registers the onset of personalization at $k \approx 2$ and the collapse at $k^\star$ that RMSE, diluted by a negative-heavy evaluation set, barely detects. At $\tau = 25$ the iALS recovery appears in NDCG and is *entirely absent* from RMSE. For Logistic MF the asymmetry reverses: RMSE traces the curve cleanly while the ranking metrics are too noisy to resolve it. This is not a contradiction. RMSE and NDCG decompose the fit differently, calibration plus ordering versus ordering alone, and which one carries the signal depends on whether the loss being minimized is itself calibrated. A probabilistic loss produces scores whose magnitudes are meaningful, so the pointwise metric is informative; squared reconstruction of a sampled support produces scores whose magnitudes are largely an artifact of the sampling ratio, so only the ordering is.

That the loss governs the metric, but not the phenomenon, is confirmed by the comparison across models, where the treatment of negatives turns out to be decisive. The two sampled-support models, NMF and Logistic MF, both exhibit double descent, at every filter and under either loss. The full-matrix model, iALS, does *not* at $\tau = 3$ and $\tau = 5$; a weak uptick appears at $\tau = 10$ and a clean, sustained

recovery only at $\tau \geq 20$. Since NMF and Logistic MF share the support but not the loss, and iALS shares neither, it is not the loss function that matters but the existence of a fixed finite support. The mechanism we propose is the one set out: a model that fits a fixed Ω has a well-defined interpolation threshold and a fixed set of noisy labels (the false negatives) to contort itself around; a model that fits all $n \times m$ cells under a permanent ℓ_2 penalty has neither, so the mechanism is correspondingly weakened and what survives of it depends on how much per-user signal the population carries. That iALS recovers double descent precisely as the population densifies is consistent with this account.

Regularization confirms the same picture without altering it. Weakening λ from 0.06 to 0.01 raises the NMF test-RMSE peak from ≈ 0.47 to ≈ 0.82 without moving its location and without altering the NDCG curve's shape. Regularization modulates the amplitude of the phenomenon; it neither creates nor destroys it, exactly as in the explicit case. The dissociation between height and location is itself informative: the location is set by parameter counting, the amplitude by how hard the model is pushed to interpolate the noisy labels that negative sampling injects.

Taken together, these results show that double descent is not a peculiarity of explicit ratings or of reconstruction error. It survives the passage to sparse, one-class, implicit-feedback e-commerce data and to ranking evaluation, but it survives *conditionally*, and the conditions are legible: a fixed finite training support with its unavoidable false negatives, enough per-user signal for an under-capacity regime to exist - which the first descent's appearance only at $\tau \geq 10$ makes vivid - weak enough regularization for the peak to show, and a metric aligned with the loss being minimized. To first approximation the peak is located by $|\Omega|/(n+m)$. Within these conditions, the conventional restriction $k \ll \min(n, m)$ is too conservative for this dataset: under NMF the best recommendations, on every metric, come from the largest model tested.

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